Fluid Flow Operation

Text Books:

- **K.A. Gavanhe, “Unit Operations - I”, Nirali Prakashan, 2015**
- W. L. McCabe, J.C. Smith and P. Harriott, “Unit operations of Chemical Engineering”, McGraw Hill, International Edn., 7th Edition, 2005.

Unit 1: Properties of fluids and concept of pressure

- Introduction
- Nature of fluids
- Physical properties of fluids
- Types of fluids
- Fluid statics: Pressure-density height relationships
- Pressure measurement
- Dimensional analysis
- Dimensionless numbers

Force

- **Newton's second law of motion: states that force (F) is proportional to the product of mass (m) and acceleration (a).**

$$F \propto m \times a$$

$$F = k \times m \times a$$

- k is a proportionality constant, in the SI system, the constant k is fixed at or taken as unity

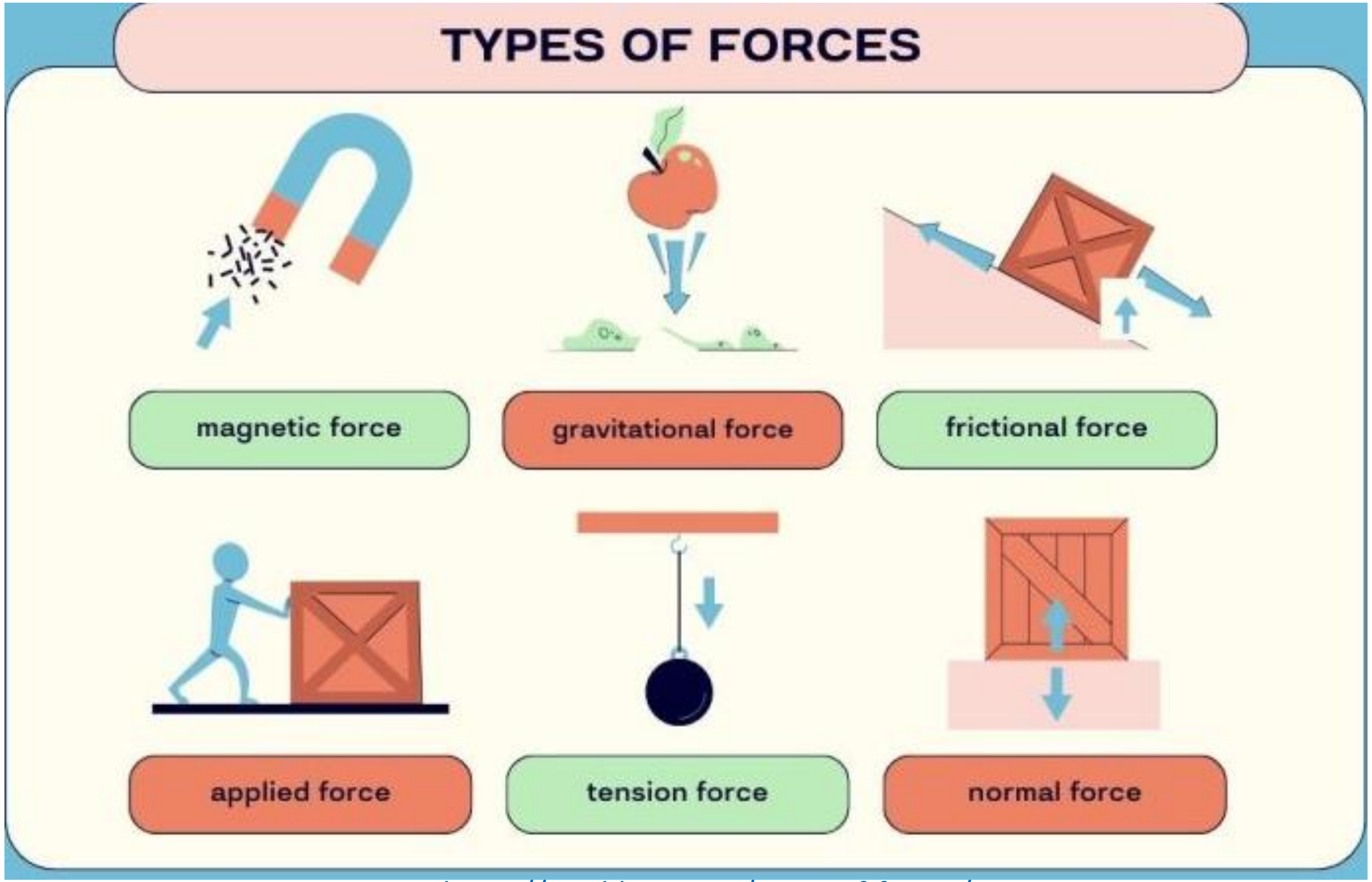
$$F = m \times a$$

- **The unit of force, newton (N) has been named after Newton.**
- Newton is the force which when applied to a body of mass 1 kg gives it an acceleration of 1 m/s^2.

$$\mathbf{1\ N = 1\ (kg.m)/s^2}$$

$$1\ N = 10^5\ dyn\ (CGS)$$

Force



https://earthhow.com/types-of-forces/

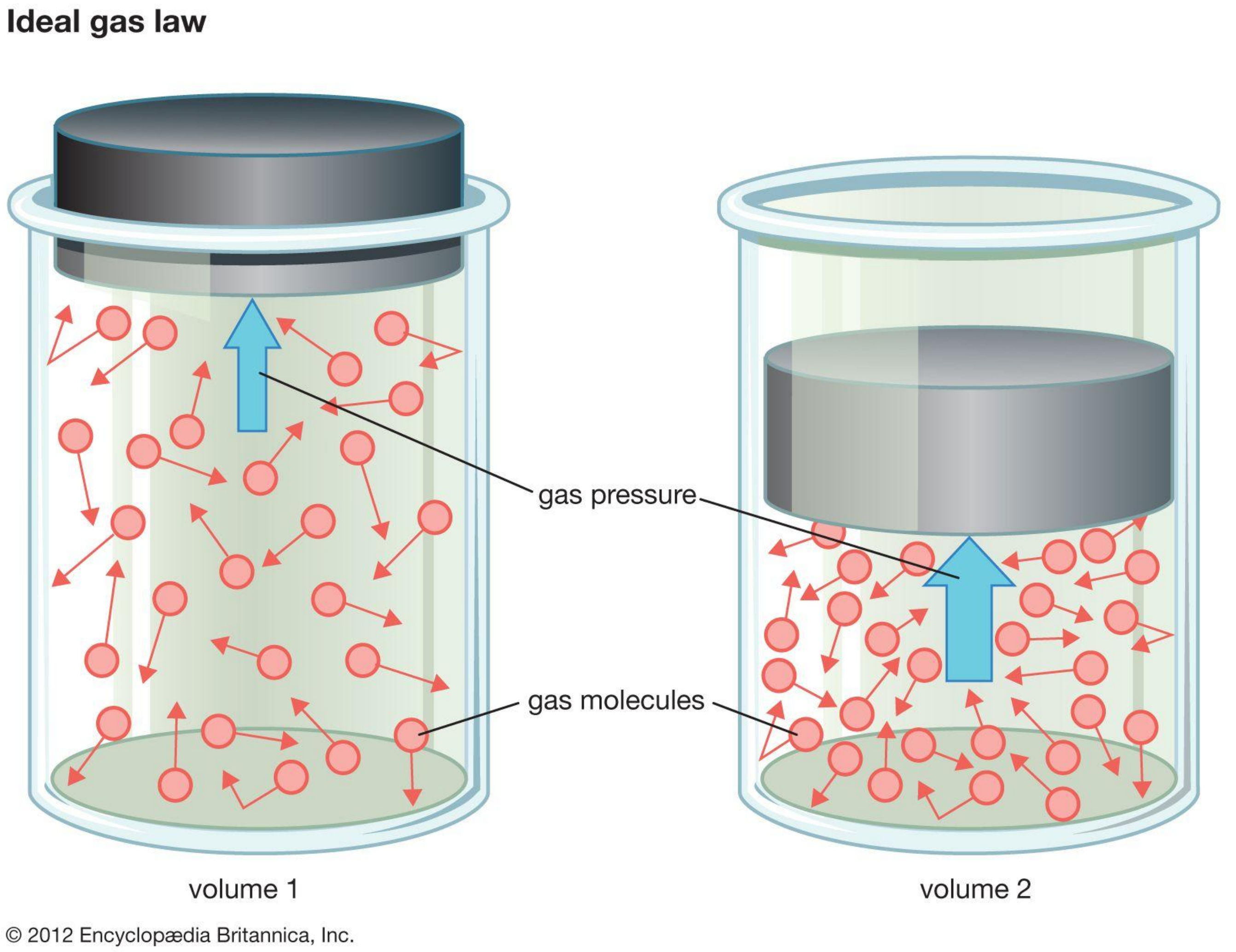
Pressure

- **Pressure is defined as the force per unit area.**
- The unit of pressure in the SI system is **newton per square meter, N/m^2**.
- In the SI system, the unit N/m^2 is called the **Pascal, Pa** (in honour of the **scientist Pascal**).
- As the unit of pressure Pa is small in magnitude, the pressure is usually expressed in SI in kilopascal (kPa).
- A multiple of pascal is called the bar and it is also used as a unit of pressure.

$$\mathbf{1\ bar = 10^5\ Pa = 10^5\ N/m}$$

- The air of the atmosphere exerts pressure on all bodies that are exposed to it.
- At the mean sea level, the pressure exerted by the air is **$101325\ N/m^2 = 101.325$ kPa**.
- This pressure is known as the **standard or normal atmospheric pressure**.

Pressure



https://www.britannica.com/science/pressure

Pressure

- **One standard atmosphere (1 atm) balances a column of 760 mm of mercury at 0°C.**

$$\mathbf{1\ atm = 760\ mm\ Hg = 101325\ Pa\ or\ N/m^2}$$

$$\mathbf{= 101.325\ kPa\ or\ kN/m^2 = 1.103\ bar = 760\ torr = 10.33\ m\ H_2O}$$

- Sub-atmospheric pressure (pressure below the actual/normal atmospheric pressure) is expressed in **torr** (in honour of the **scientist Torricelli**), and is used for systems under **vacuum**.
- Pressure is measured with the help of a **pressure gauge**.
- ***The pressure gauge registers the difference between the pressure prevailing in the vessel (actual pressure) and the local atmospheric pressure.***
- The pressure registered by the pressure gauge is called the **gauge pressure** and therefore the letter 'g' follows the unit of pressure [2 atm_g, 300 kPa_g].

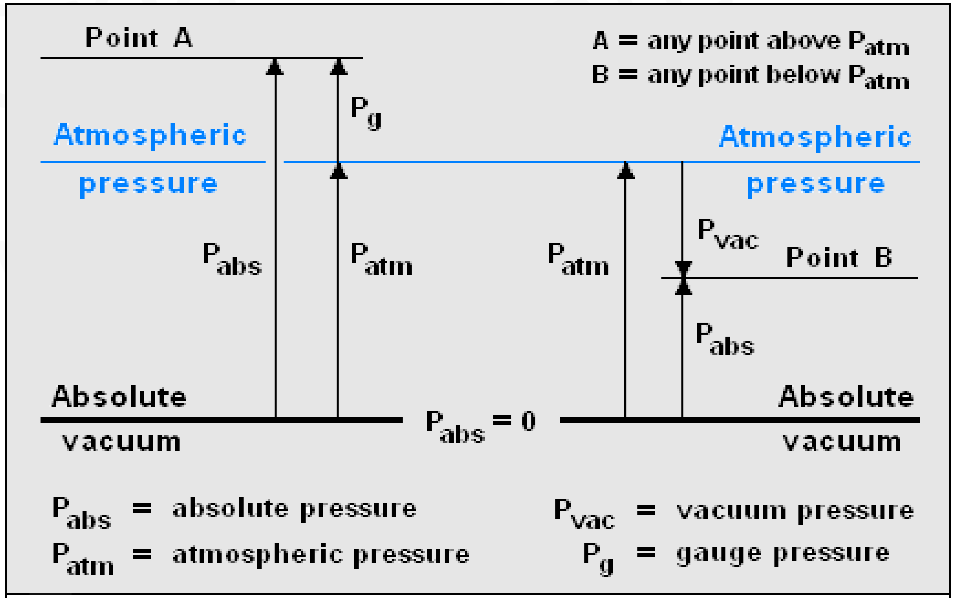
Pressure

- If no letter follows the pressure units, it is taken as absolute pressure.

Absolute pressure = gauge pressure + atmospheric pressure

Absolute pressure = atmospheric pressure – vacuum

https://www.processparameters.co.uk/pressure-measurement/



The relationship between absolute, gauge and vacuum pressures.

https://instrumentationtools.com/difference-between-absolute-and-gauge-pressure/

Work/ Energy

- **Work (energy) is defined as the product of the force acting on a body and the distance travelled by the body in the direction of the applied force.**
- If force F acting on a body moves it through a distance L, then the work done on the body is given by

$$\mathbf{W = F \times L \Rightarrow Newton \times metre \Rightarrow N.m \Rightarrow joule\ (J)}$$

- The unit of work (energy) in the SI system is **newton-metre (N.m)** or joule (abbreviated as J).
- **Energy is defined as the capacity of a body for doing work, mechanical work can produce a property of the matter called energy.**
- Energy is present in a system in different forms, e.g., mechanical-potential, kinetic, chemical, thermal and electrical.
- Any form of energy can be converted into work, so the units of work and energy are the same.

Power

- **Power is defined as the work done per unit time or the rate at which work is done.**

$$\mathbf{P = W \times t \Rightarrow joule\ second \Rightarrow Watt}$$

- The **SI unit of power is watt, abbreviated as W.**
- Watt is the power that gives rise to the production of energy at the rate of one joule per second.

$$1\ W = 1\ J/s$$

$$1\ \text{Metric horse power (hp)} = 735.5\ W$$

Units and dimensions

Basic Quantity	System of Units				Dimension
	CGS	MKS	FPS	SI	
Length	Centimetre (cm)	Metre (m)	Foot (ft)	Metre (m)	L
Mass	Gram (gm)	Kilogram (kg)	Pound (*l*b)	Kilogram (kg)	M
Temperature	Celsius (^oC)	Celsius (^oC)	Fahrenheit (^oF)	Kelvin (K)	T
Time	Second (s)	Second (s)	Second (s)	Second (s)	θ

Derived Quantities

Derived Quantity	Units in SI system	Abbreviated units	Dimension
Area	square meters	m^2	L^2
Volume	cubic meters	m^3	L^3
Density	kilograms per cubic meter	kg/m^3	ML^{-3}
Linear velocity	meters per second	m/s	$L\theta^{-1}$
Linear acceleration	meters per second per second	m/s^2	$L\theta^{-2}$
Capacity	cubic meters	m^3	L^3
Force	newtons	N	$ML\theta^{-2}$
Pressure	newtons per square meter (pascal)	N/m^2 (Pa)	$ML^{-1}\theta^{-2}$

Derived Quantities

Pressure	newtons per square meter (pascal)	N/m^2 (Pa)	$ML^{-1}\theta^{-2}$
Mass flow rate	kilograms per hour	kg/h	$M\theta^{-1}$
Volumetric flow rate	cubic meters per hour	m^3/h	$L^3\theta^{-1}$
Work/energy	joules	J	$ML^2\theta^{-2}$
Heat/Enthalpy	joules	J	$ML^2\theta^{-2}$
Power	kilowatts	kW	$ML^2\theta^{-3}$
Viscosity	newtons second per square meter [kilograms per meter per second]	$(N\cdot s)/m^2$ or $Pa\cdot s$ [$kg/(m\cdot s)$]	$ML^{-1}\theta^{-1}$

Fluids

- **A fluid is a substance which is capable of flowing if allowed to do so.**
- **A fluid is a substance which undergoes continuous deformation when subjected to shear force.**
- **Gases, vapours and liquids** possessing the above characteristics are referred to as fluids.
- Handling fluids is much simpler, cheaper and less difficult than handling of solids.
- Therefore, whenever possible, it is desirable to handle everything in the form of liquids, solutions or suspensions.

Fluid Flow Operations

- **The branch of engineering science which deals with the behaviour of fluids at rest or in motion is called FLUID MECHANICS.**
- The study of water is referred to as Hydraulics.
- Fluid mechanics is classified as :

1. **Fluid statics** deals with the **study of fluids at rest** which involves the study of pressure exerted by a fluid at rest and the variation of fluid pressure throughout the fluid.
2. **Fluid dynamics** deals with the **study of fluids in motion** relative to stationary solid walls or boundaries.

- Transportation of fluids (liquids and gases) from one location to another through pipes or ducts requires determination of the pressure drop through the system and of the power required for pumping, selection of a suitable type of pumping device and measurement of the flow rates.

Nature of fluid

1. A fluid is a substance which is capable of flowing if allowed to do so.
2. A fluid is a substance that has no definite shape of its own, but conforms to the shape of the containing vessel.
3. A fluid is a substance which undergoes continuous deformation when subjected to a shearing force/shear force.

- Since liquids and gases / vapours possess the above cited characteristics, they are referred to as fluids.

Properties of fluids

1. Mass density (specific mass) or density (ρ)
2. Weight density (specific weight) (w)
3. Vapour pressure
4. Specific gravity
5. Viscosity
6. Surface tension and capillarity
7. Compressibility and elasticity
8. Thermal conductivity
9. Specific volume

Properties of fluids

- **Density :** Density (ρ) or mass density of a fluid is the mass of the fluid per unit volume.
- $\boldsymbol{\rho = \frac{mass}{volume} = \frac{kg}{m^3}}$
- In the SI system, it is expressed in kg/m^3.
- The density of pure water at 277 K (4 ^oC) is taken as 1000 kg/m^3.

- **Weight Density :** Weight density of a fluid is the weight of the fluid per unit volume.
- In the SI system, it is expressed in N/m^3, where N =$kg.m/s^2$.
- The relation between mass density and weight density is $\mathbf{w} = \boldsymbol{\rho g}$ where g is the acceleration due to gravity (9.81 m/s^2).
- Specific weight or weight density of pure water at 277 K (4 ^oC) is taken as 9810 N/m^3.

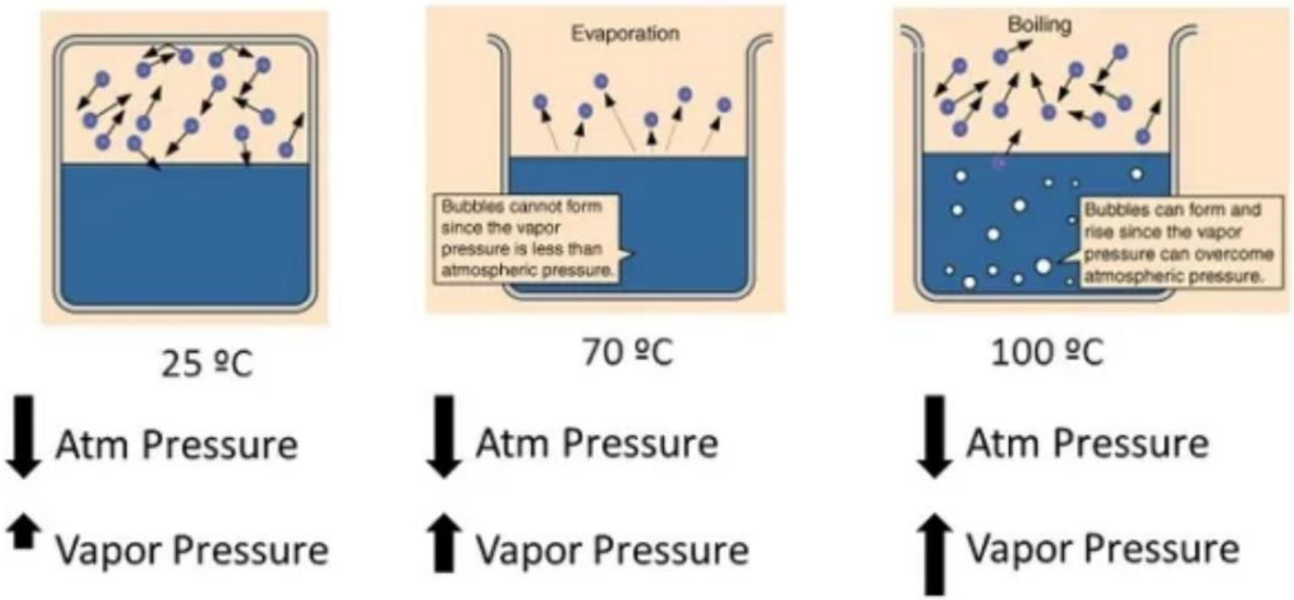
Properties of fluids

- **Specific Volume :** Specific volume of a fluid is the volume of the fluid per unit mass.
- $\boldsymbol{\gamma = \frac{volume}{mass} = \frac{1}{\rho} = \frac{m^3}{kg}}$
- In the SI system, it is expressed in m^3/kg.

- **Specific Gravity :** The specific gravity of a fluid is the ratio of the density of the fluid to the density of a standard fluid.
- $\boldsymbol{\gamma_f = \frac{\rho_{fluid}}{\rho_{standard\ fluid}} = \frac{\rho_{fluid}}{\rho_{water\ at\ 4\,^oC}}}$
- For liquids, water at 277 K (4 ^{o}C) is considered as a standard fluid and for gases, air at NTP (0°C and 760 torr) is considered as a standard fluid.

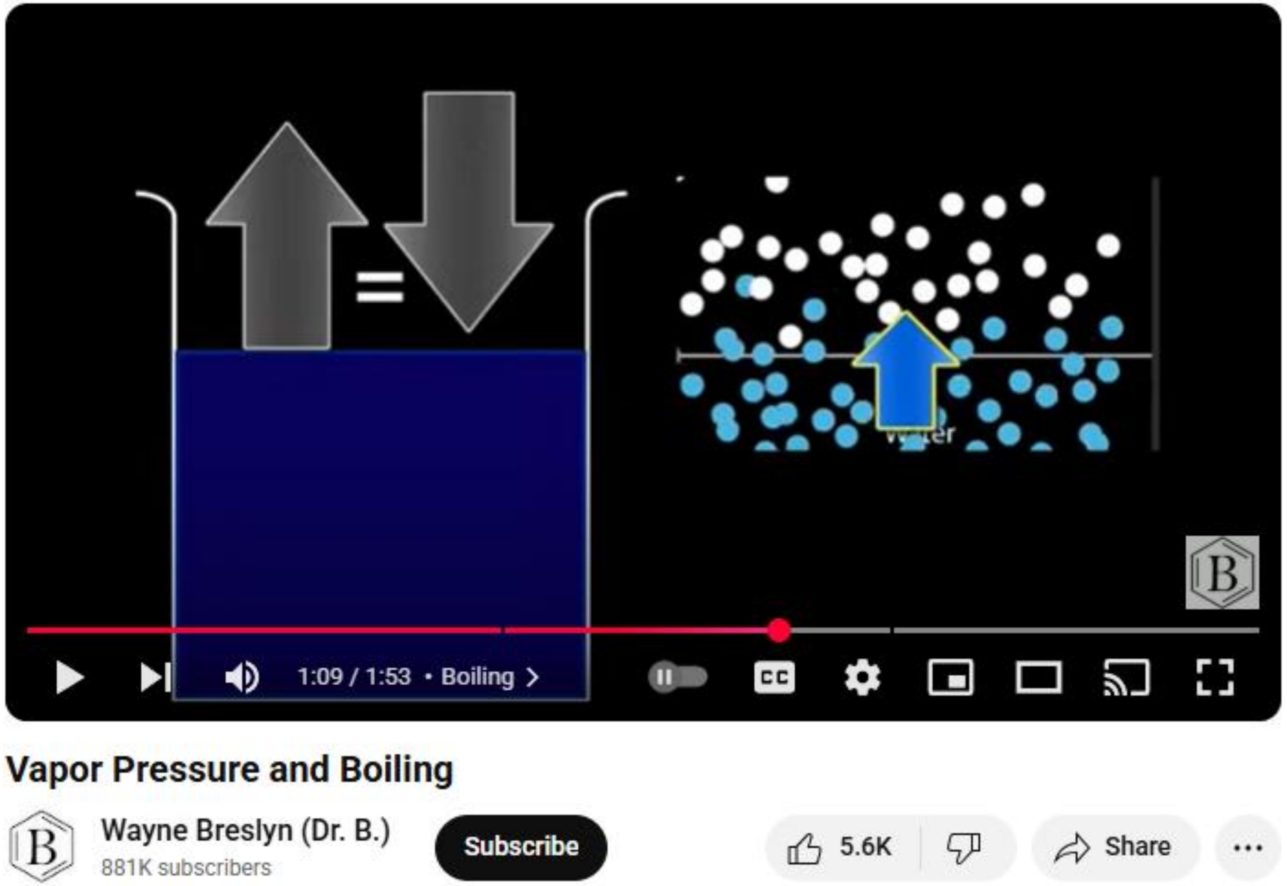
Properties of fluids

- **Vapour Pressure :**
- **The vapour pressure of a pure liquid is defined as the absolute pressure at which the liquid and its vapour are in equilibrium at a given temperature or pressure.**
- The pressure exerted by the vapour (on the surface of a liquid) at equilibrium conditions is called as the vapour pressure of the liquid at a given temperature.
- Pure air free water exerts a vapour pressure of 101.325 kPa (760 torr) at 373.15 K (100 ^{o}C).



https://cncontrolvalve.com/vapor-pressure-the-science-applications-and-importance-explained/

Vapor pressure



https://www.youtube.com/watch?v=ffBusZO-TO0

Properties of fluids

- **Surface Tension : The property of liquid surface film to exert tension is called as the surface tension.**
- It is the force required to maintain a unit length of film in equilibrium.
- $\sigma = \frac{F}{L} = \frac{N}{m}$

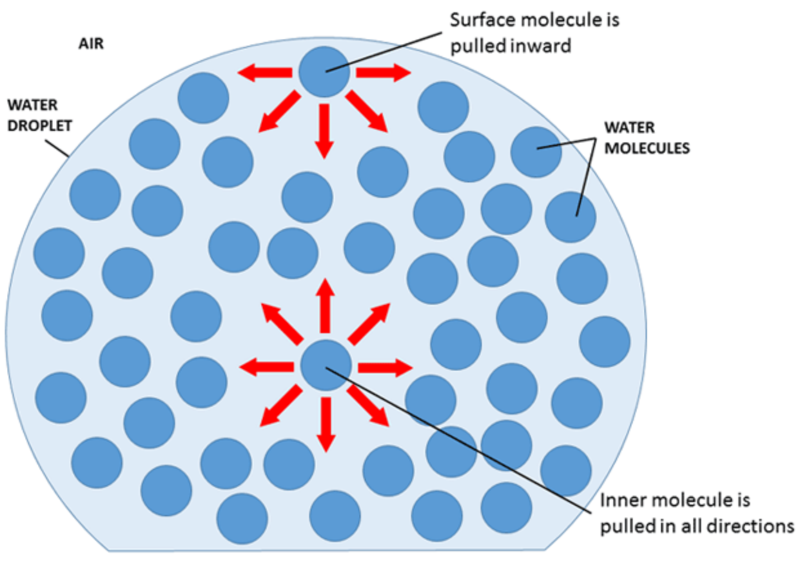
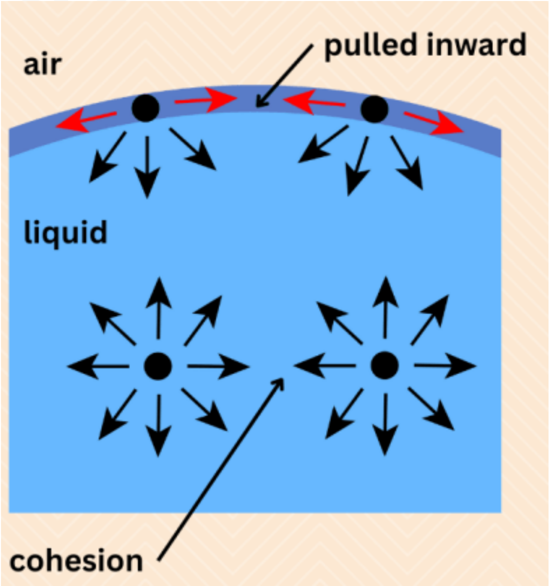


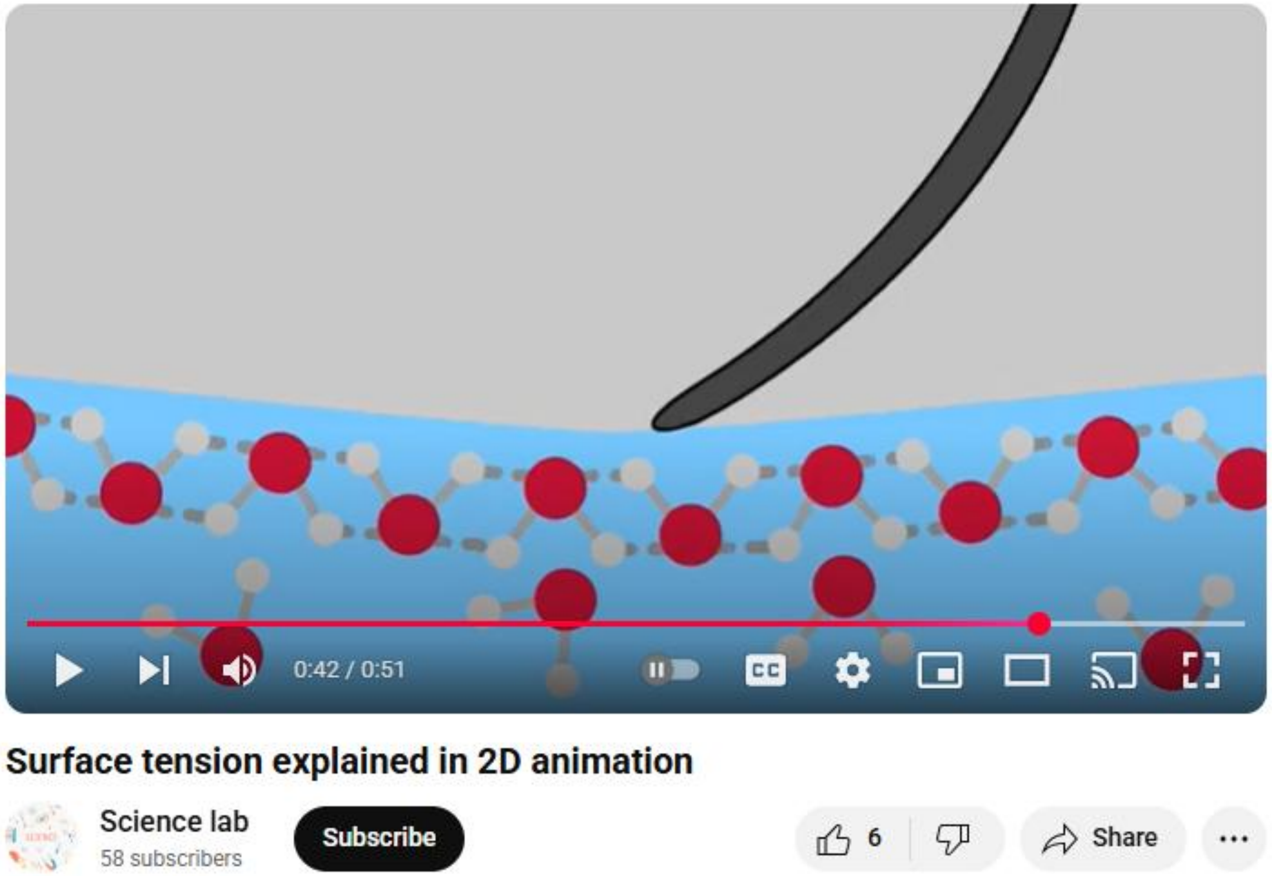
Diagram not to scale

https://www.sciencebuddies.org/science-fair-projects/project-ideas/Chem_p021/chemistry/measuring-surface-tension-of-water-with-a-penny



https://sciencenotes.org/surface-tension-definition-examples-formula/

Surface Tension



https://www.youtube.com/watch?v=ws0jz_HY1ow

Surface Tension

Liquid	Surface Tension γ (N/m)
Water at 0 °C	0.0756
Water at 20 °C	0.0728
Water at 100 °C	0.0589
Soapy water (typical)	0.0370
Ethyl alcohol	0.0223
Mercury	0.465
Olive oil	0.032
Tissue fluids (typical)	0.050
Blood, whole at 37 °C	0.058
Blood plasma at 37 °C	0.073
Gold at 1070 °C	1.000
Benzene	0.0282
Oxygen at −193 °C	0.0157
Helium at −269 °C	0.00012

https://www.researchgate.net/publication/385674236_The_Effect_of_Carbon_Nanotubes_on_the_Viscosity_and_Surface_Tension_of_Heat_Transfer_Fluids-A_Review_Paper

Table 2. Surface Tensions of Common Substances at 25 °C

Substance	Formula	Surface Tension (mN/m)
water	H_2O	71.99
mercury	Hg	458.48
ethanol	C_2H_5OH	21.97
octane	C_8H_{18}	21.14
ethylene glycol	$CH_2(OH)CH_2(OH)$	47.99

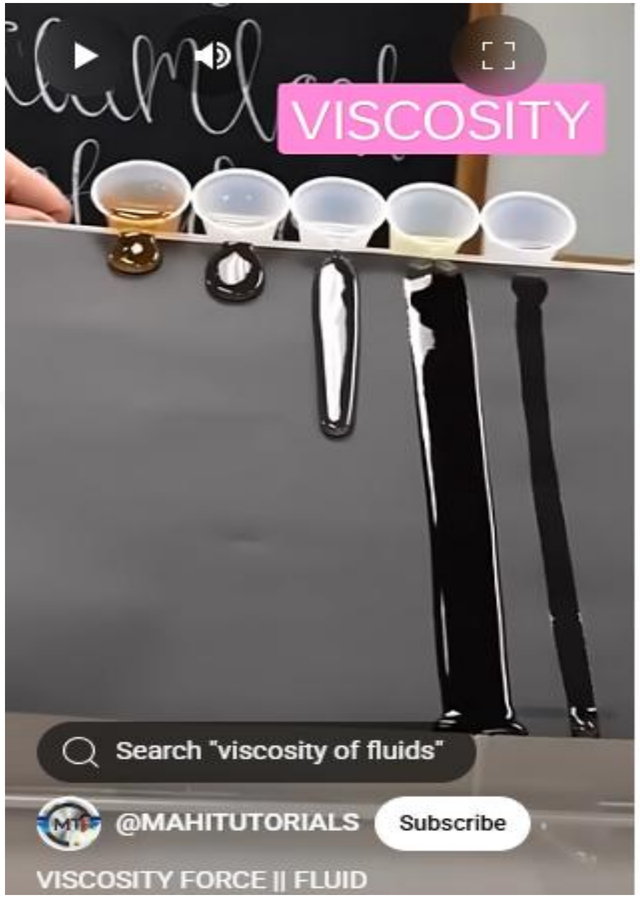
https://courses.lumenlearning.com/suny-chem-atoms-first/chapter/properties-of-liquids/#:~:text=Water%2C%20gasoline%2C%20and%20other%20liquids,Figure%201%2C%20have%20higher%20viscosities.

Properties of fluids

- **Viscosity :**
- A fluid undergoes continuous deformation when subjected to a shear stress.
- **The resistance offered by a fluid to its continuous deformation (when subjected to a shear stress/force) is called viscosity**
- The viscosity of a fluid at a given temperature is a **measure of its resistance to flow**.
- The viscosity of a fluid (gas or liquid) is practically independent of the pressure but varies with temperature.
- For gases, viscosity increases with an increase in temperature, while for liquids it decreases with an increase in temperature.

Viscosity

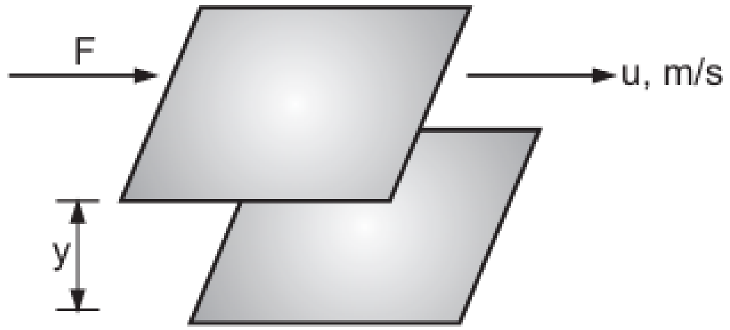
https://sciencequery.com/high-viscosity-vs-low-viscosity/



https://youtube.com/shorts/sDFiWcvNCiI?feature=shared

Interesting Video to understand viscosity:
https://www.youtube.com/watch?v=aGxR6pj8E0A

Properties of fluids



- **Viscosity:**
- Consider two layers of a fluid are 'y' cm apart, area of each of these layers be A cm^2.
- The top layer is moving parallel to the bottom layer at a velocity of 'u' cm/s relative to the bottom layer.
- To maintain this motion, i.e., the velocity 'u' and to overcome the fluid friction between these layers, for any actual fluid, a force of 'F' is required.
- Experimentally $F \propto u.A/y$

Keeping proportionality constant μ

$$F = \mu\, u\, A/y$$

$$F/A = \mu.u/y$$

- Shear stress, τ $\tau = F/A = \mu.u/y$

Differential form

Newton's Law of Viscosity

$$\boxed{\tau = \mu \cdot \frac{du}{dy}}$$

Properties of fluids

- **Coefficient of viscosity/ dynamic viscosity (involves force)/ viscosity of a fluid μ**

$$\mu = \frac{\tau}{du/dy} \quad = \frac{N/m^2}{m/m/s} = \frac{N.s}{m^2} = Pa.s = \frac{kg}{m.s}$$

- **Dynamic viscosity μ, may be defined as the shear stress required to produce unit rate of shear deformation (or shear rate).**
- As the unit $(N.s)/m^2$ is very large for most of the fluids, it is customary to express viscosity as $(mN.s)/m^2$ or mPa.s
- In the C.G.S. system, viscosity may be expressed in poise (P) (the unit poise is named after the French scientist Poiseuille) or centipoise (cP).

$$1 \text{ poise} = 1\text{ P} = 1\text{ gm/(cm.s)} = 0.10\text{ kg/(m.s)} = 0.10\text{ (N.s)/m}^2 \text{ or Pa.s} = 100\text{ cP}$$

Properties of fluids

- **The kinematic viscosity υ of a fluid is defined as the ratio of the viscosity of the fluid to its density.**

$$\upsilon = \mu/\rho = \frac{m^2}{s}$$

- The C.G.S. unit of kinematic viscosity is termed as stoke and is equal to 1 cm^2/s.

Viscosity

Table 1. Viscosities of Common Substances at 25 °C

Substance	Formula	Viscosity (mPa·s)
water	H_2O	0.890
mercury	Hg	1.526
ethanol	C_2H_5OH	1.074
octane	C_8H_{18}	0.508
ethylene glycol	$CH_2(OH)CH_2(OH)$	16.1
honey	variable	~2,000–10,000
motor oil	variable	~50–500

https://courses.lumenlearning.com/suny-chem-atoms-first/chapter/properties-of-liquids/#:~:text=Water%2C%20gasoline%2C%20and%20other%20liquids,Figure%201%2C%20have%20higher%20viscosities.

Viscosity

Type of Liquid	Centipoise (cp)	Centistokes (cSt)
Water	1	1
Milk	3	4
Blood or Kerosene	10	
No. 4 Fuel Oil	12.6	15.7
Anti-Freeze or Ethylene Glycol	15	
Cream	20	20.6
Vegetable Oil	40	43.2
SAE 10 Oil	88	110
Tomato Juice	180	220
SAE 30	352	440
Glycerin	800	1,100
Honey	1,500	2,200
Glue	3,000	4,500
Mayonnaise	5,000	6,250
Molasses	8,640	10,800
Sour Cream	15,000	19,000
SAE 70 Oil	17,640	19,600
Window Putty	100,000	

https://www.epakmachinery.com/chart-viscosity-of-common-liquids/

Types of fluid

Ideal Fluid:

- It is a fluid which does not offer resistance to flow / deformation / change in shape, i.e., it has no viscosity.
- It is frictionless and incompressible.
- However, an ideal fluid does not exist in nature and therefore, it is only an imaginary fluid.

Real Fluid:

- It is a fluid which offers resistance when it is set in motion.
- All naturally occurring fluids are real fluids.

Pressure

- **The basic property of a static fluid is pressure.**
- When a certain mass of fluid is contained in a vessel, it exerts forces at all points on the surfaces of the vessel in contact.
- The forces exerted always act in the direction normal to the surface in contact.
- **The normal force exerted by a fluid per unit area of the surface is called as the fluid pressure.**
- **If F is the force acting on the area A, then the pressure or intensity of pressure is given by P = F/A.**
- In a static fluid, the pressure at any given point is the same in all the directions.
- ***If the pressure at a given point was not the same in all directions, there would be non-equilibrium and the resultant force should exist. As the fluid is in static equilibrium, there is no net unbalanced force at any point.***

Pressure

Forces on static fluid element:

1. Force of gravity acting downward
2. Pressure force acting upward
3. Vertical component of pressure force acting downward

- Since the fluid is in equilibrium, resultant of forces is zero
- Force balance parallel to x, y, and z axis respectively show

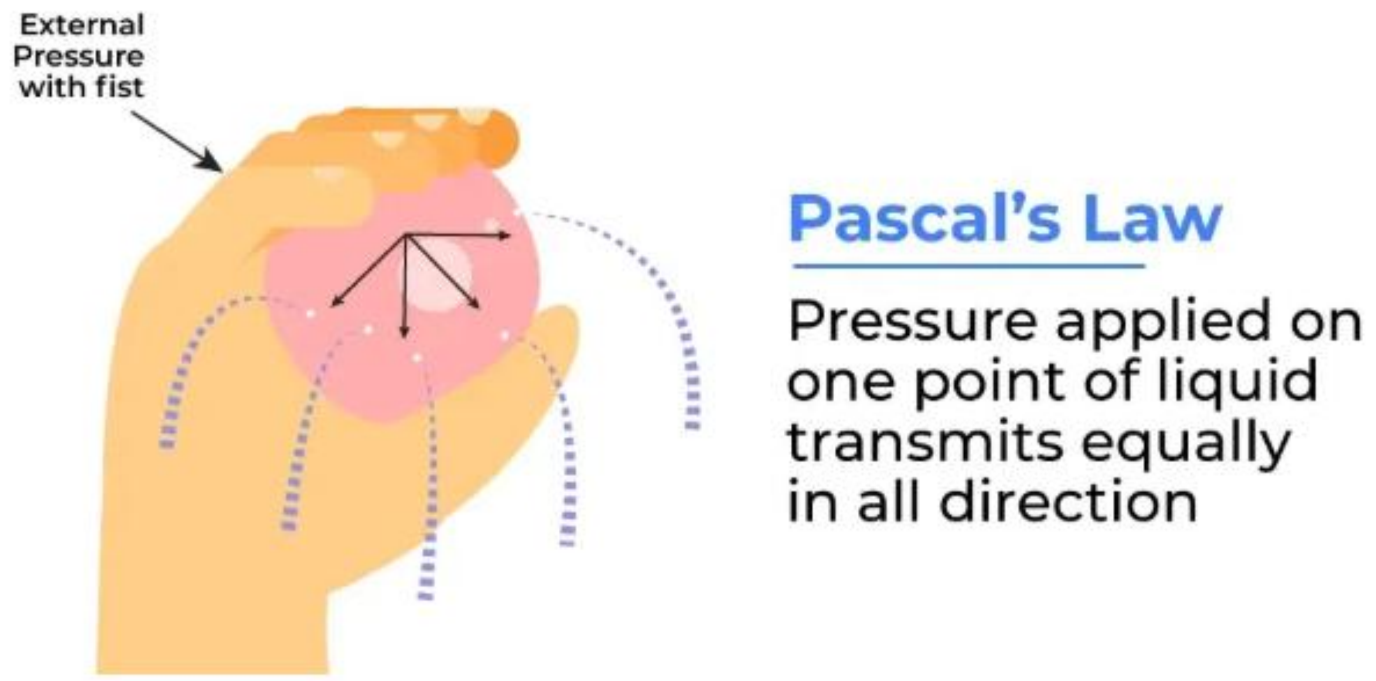
$$p_x = p_y = p_z = p$$

- This is known as Pascal's principle.
- **Pascal discovered that the pressure at a point in a fluid at rest is the same in all directions.**

W. L. McCabe, J.C. Smith and P. Harriott, "Unit operations of Chemical Engineering", McGraw Hill, International Edn., 7th Edition, 2005.

Pressure

- **Pascal’s principle:** Pascal's law states that whenever external pressure is applied to any part of a liquid. This pressure gets distributed in all directions equally.

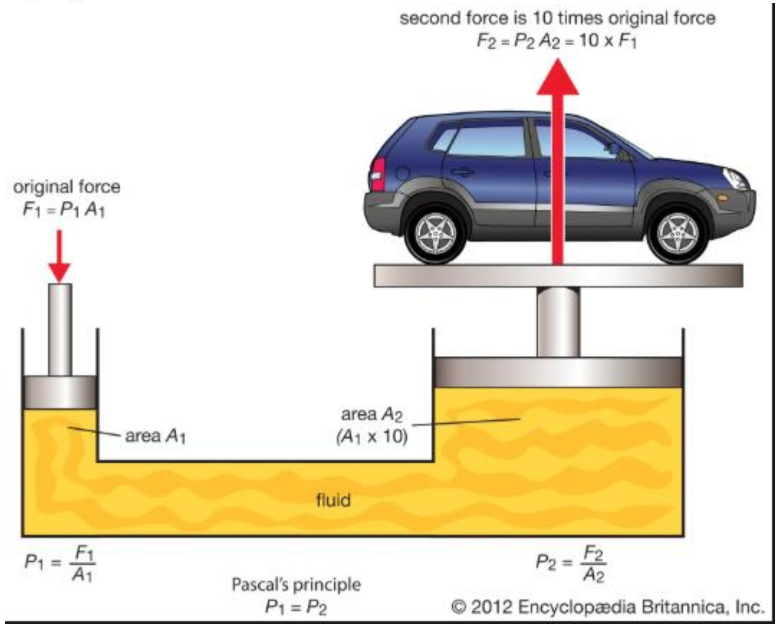


https://www.geeksforgeeks.org/physics/pascals-law/

Pressure

- **Pascal's principle:** in a fluid at rest in a closed container, a pressure change in one part is transmitted without loss to every portion of the fluid and to the walls of the container.
- The principle was first enunciated by the **French scientist Blaise Pascal**.
- According to Pascal's principle, in a hydraulic system a pressure exerted on a piston produces an equal increase in pressure on another piston in the system.

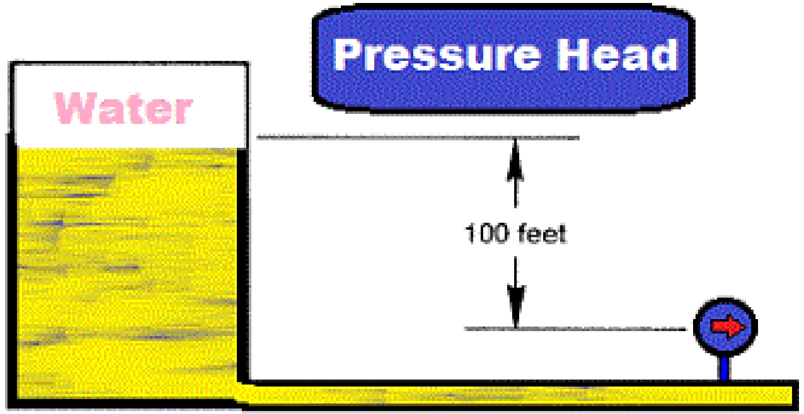
- If the second piston has an area 10 times that of the first, the force on the second piston is 10 times greater, though the pressure is the same as that on the first piston.
- This effect is exemplified by the hydraulic press, based on Pascal's principle, which is used in such applications as hydraulic brakes.



https://www.britannica.com/science/Pascals-principle

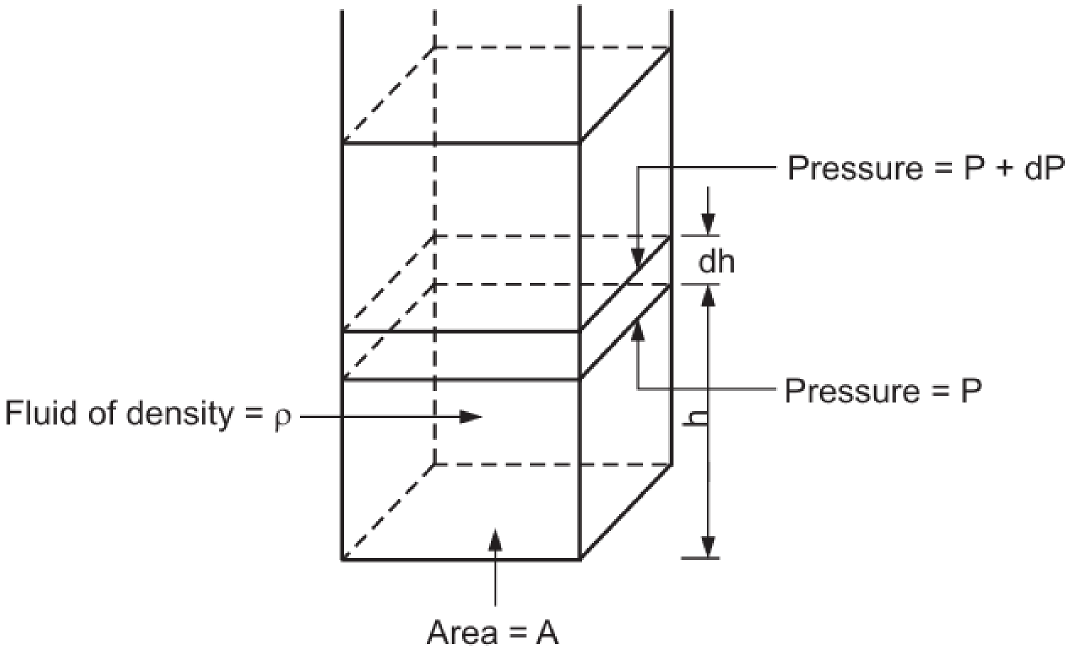
Pressure Head

- **As the pressure at any point in a static liquid depends upon the height of the free surface above the point, it is convenient to express a fluid pressure in terms of pressure head.**
- The pressure head is then expressed in terms of meters of a liquid column.
- The vertical height or the free surface above any point in a liquid at rest is called as the pressure head.



$$h = \frac{P}{\rho g}, \quad \frac{N/m^2}{kg/m^3 \times m/s^2} = \frac{(kg.m/s^2)/m^2}{(kg/m^3)\ (m/s^2)} = m$$

Hydrostatic equilibrium



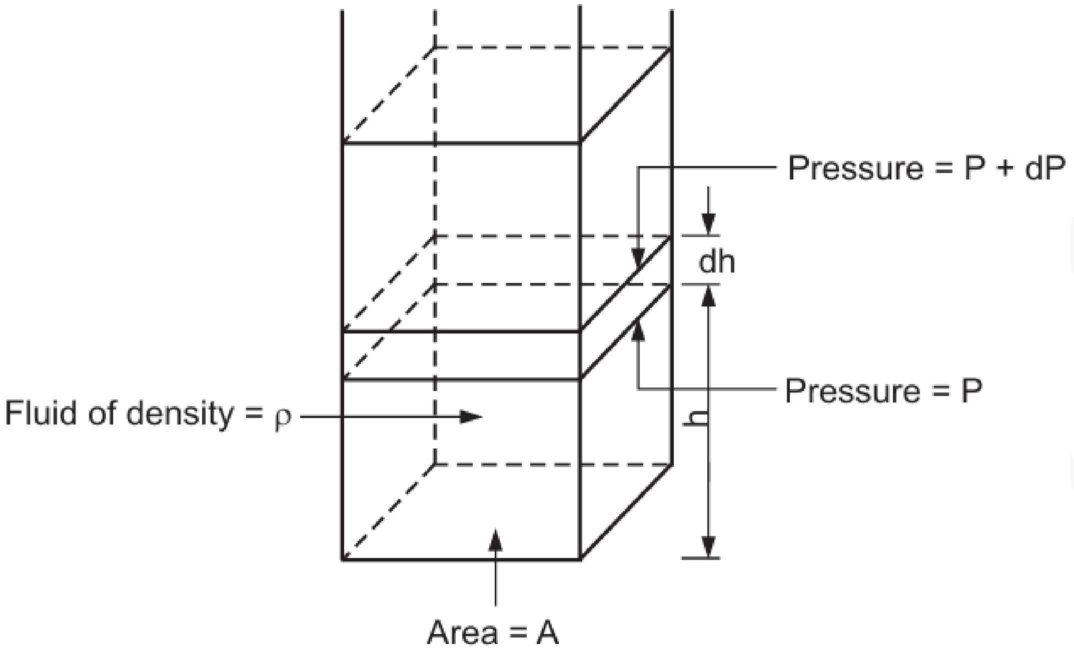
- In a vertical column of the static fluid, the pressure at any point is the same in all directions.
- Pressure varies with the height of the column.

Force balance on small fluid element of height dh:

1. (P + dP)A (acting downward, +ve)
2. PA (acting upward, -ve)
3. Force due to gravity = mg = Vρg = A.dh.ρ.g (-ve)

- *As the fluid element is in equilibrium, the resultant of these three forces acting on it must be zero.*

Hydrostatic equilibrium



$$+ P.A - A.\, dh.\rho.g - (P + dP)\, A = 0$$

$$P.A - A.\, dh.\rho.g - PA - A.dP = 0$$
$$- A.dh.\rho.g - A.dP = 0$$
$$dP + dh.\rho.g = 0$$

Hydrostatic equilibrium – Incompressible fluids

- **For incompressible fluids, density is independent of pressure.**
- Integrating

$$dP + g.\rho.dh = 0$$

$$\int dP + g.\rho.\int dh = 0$$

- If the pressure at the base of the column is P_1 where h = 0 and the pressure at any height h above the base is P_2 such that $P_1 > P_2$, then

$$\int_{P_1}^{P_2} dP = -g.\rho \int_0^h dh$$

$$(P_1 - P_2) = h.\rho.g \qquad \boxed{\Delta P = \rho g h}$$

- Pressure is maximum at the base of the column or container of the fluid and it decreases as we move up the column.

Hydrostatic equilibrium – Compressible fluids

- **For compressible fluids, density varies with pressure.**
- For an ideal gas, the density is given by the relation: $\rho = \dfrac{PM}{RT}$

M = molecular weight of gas
R = universal gas constant
T = absolute temperature.

$$dP + dh.\rho.g = 0$$

$$dP + g\,(PM/RT)\,dh = 0$$

$$\frac{dP}{P} + g \cdot \frac{M}{RT}\,dh = 0$$

$$\ln\frac{P_2}{P_1} = -\,g \cdot \frac{M\,(h_2 - h_1)}{RT} \longrightarrow$$

Barometric equation

$$\boxed{\frac{P_2}{P_1} = \exp\cdot\left[-\,g \cdot \frac{M}{RT}(h_2 - h_1)\right]}$$

Barometric equation gives pressure distribution within an ideal gas for isothermal conditions.

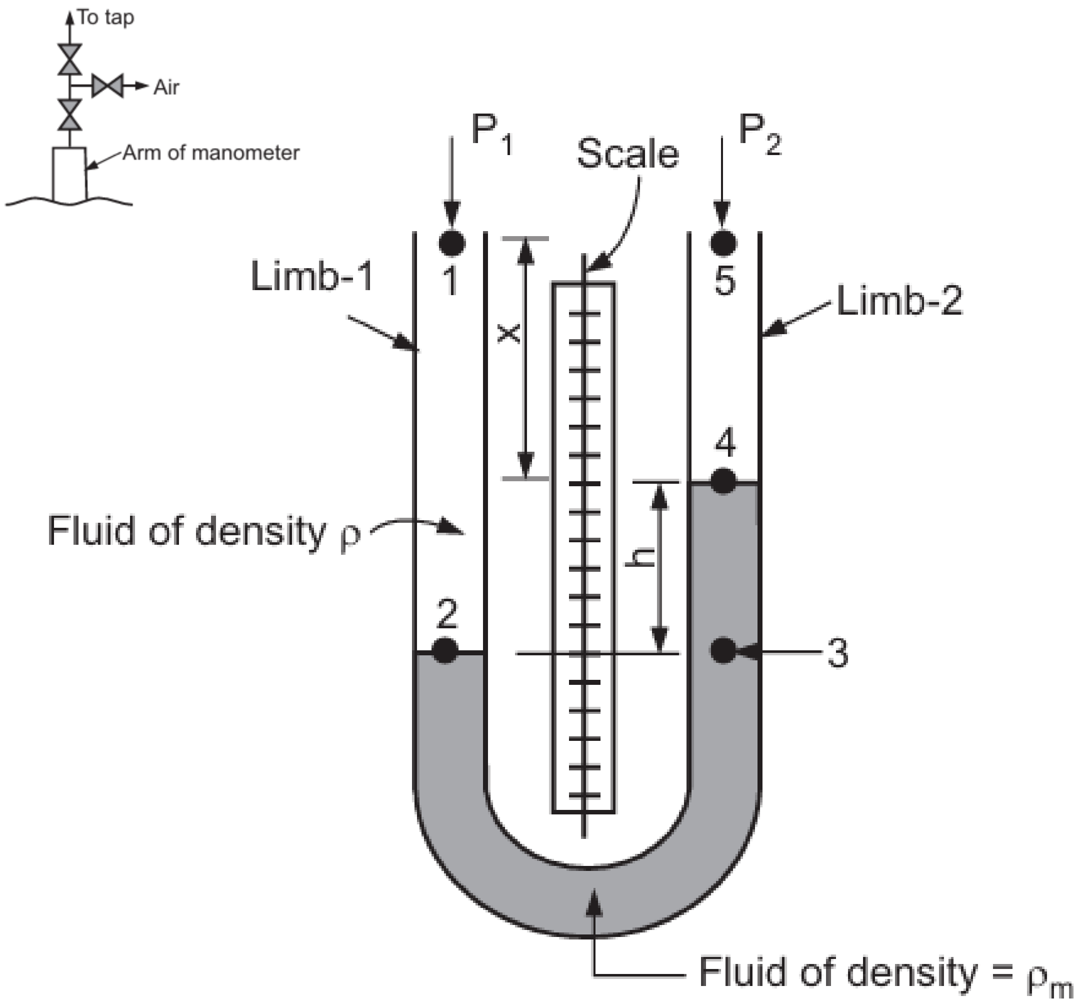
Pressure measurement: U-tube Manometer

- **Manometers are the simplest pressure measuring devices and are used for measuring low pressure or pressure differences.**

U-tube Manometer:

- U-tube manometer is the simplest form of manometer.
- **It consists of a small diameter U-shaped tube of glass.**
- The tube is clamped on a wooden board.
- **Between the two arms or legs of the manometer, a scale is fixed on the same board.**
- **The U tube is partially filled with a manometric fluid which is heavier than the process fluid.**
- The two limbs of the manometer are connected by a tubing to the taps between which the pressure drop is to be measured.

Pressure measurement: U-tube Manometer



https://www.3bscientific.com/in/u-tube-manometer-s-1000792-u8410450-3b-scientific,p_856_14309.html

https://www.indiamart.com/proddetail/u-tube-manometer-15286061662.html

Pressure measurement: U-tube Manometer

- Air vent valves are provided at end of each arm for the removal of trapped air in arm.
- **The manometric fluid is immiscible with the process fluid. The common manometric fluid is mercury**.
- U-tube manometer is filled with a given manometric fluid (fluid M) upto a certain height.
- The remaining portion of the U-tube is filled with the process fluid/flowing fluid of density ρ including the tubings.
- **One limb of the manometer is connected to the upstream tap in a pipeline and the other limb is connected to the downstream tap in the pipeline between which the pressure difference $P_1 – P_2$ is required to be measured.**
- Air, if any, is there in the line connecting taps and manometer is removed.
- **At steady state, for a given flow rate, the reading of the manometer: the difference in the level of the manometric fluid in the two arms is measured** and it gives the value of pressure difference in terms of manometric fluid across the taps (stations).

Pressure measurement: U-tube Manometer

- Consider a U-tube manometer connected in a pipeline.
- **Let pressure P_1 be exerted in one limb of the manometer and pressure P_2 be exerted in the other limb of the manometer.**
- **If P_1 is greater than P_2, the interface between the two liquids in the limb 1 will be depressed by a distance 'h' below that in the limb 2.**
- To arrive at a relationship between the pressure difference ($P_1 – P_2$) and the difference in the level in the two limbs of the manometer in terms of manometric fluid (h), pressures at points 1, 2, 3, 4, and 5 are considered.

Pressure at point 1 = P_1

Pressure at point 2 = $P_1 + (x + h)\,\rho.g$

Pressure at point 3 = Pressure at point 2 = $P_1 + (x + h)\,\rho.g$

(as the points 2 and 3 are at the same horizontal plane)

Pressure measurement: U-tube Manometer

Pressure at point 4 = $P_1 + (x + h)\,\rho.g - h\,.\,\rho_M.g$

Pressure at point 5 = $P_1 + (x + h)\,\rho.g - h\,.\,\rho_M.g - x.\rho.g$

Pressure at point 5 = P_2

$$P_2 = P_1 + (x + h)\,.\,\rho.g - h\,\rho_M.g - x\,.\,\rho.g$$

$$\mathbf{P_1 - P_2 = \Delta P = h\,(\rho_M - \rho)g}$$

where ΔP is the pressure difference and 'h' is the difference in levels in the two arms of the manometer in terms of manometric fluid.

If the flowing fluid is a gas, density ρ of the gas will normally be small compared with the density of the manometric fluid, ρ_M and thus Equation reduces to

$$\mathbf{\Delta P = P_1 - P_2 = h\,.\,\rho_M.g}$$

Pressure measurement: Inclined Manometer

- **Inclined manometers are used for measuring small pressure differences.**
- One arm of the manometer is inclined at an **angle of 5 to 10°** with the horizontal so as to obtain a larger reading.
- E.g., movement of 7 to 10 mm is obtained for a pressure change corresponding to 1 mm head of liquid.

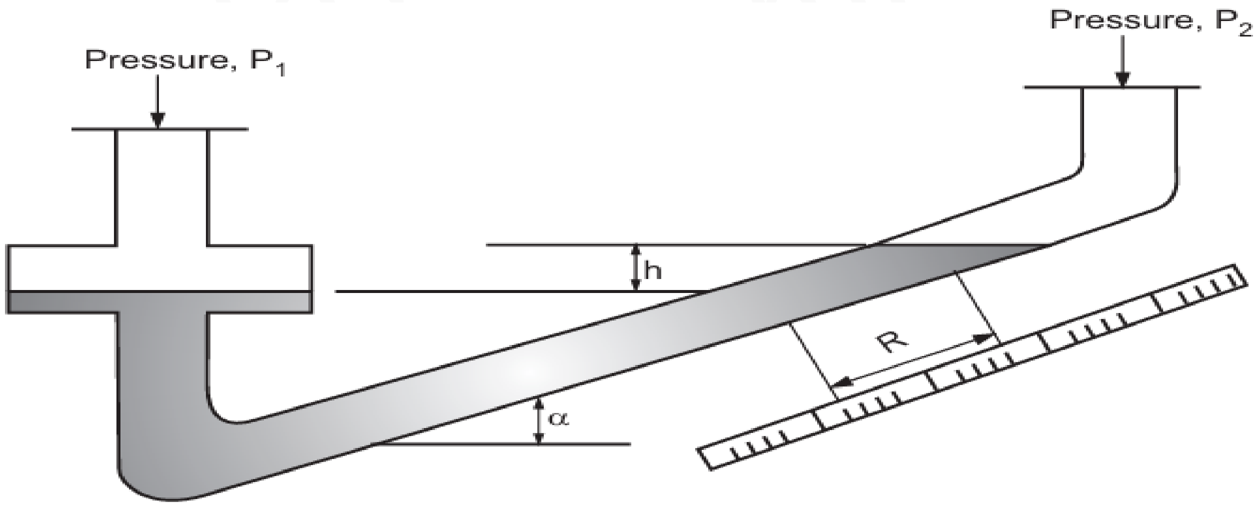


Fig. 7.5 : Inclined Tube Manometer

Pressure measurement: Inclined Manometer

- In the vertical leg of this manometer an enlargement is provided so that the movement of the meniscus in this enlargement is negligible within the operating range of the manometer.
- If the reading R (in m) is taken as distance travelled by the meniscus of the manometric fluid along the tube, then

$$\mathbf{h = R \sin \alpha}$$

where α = angle of inclination

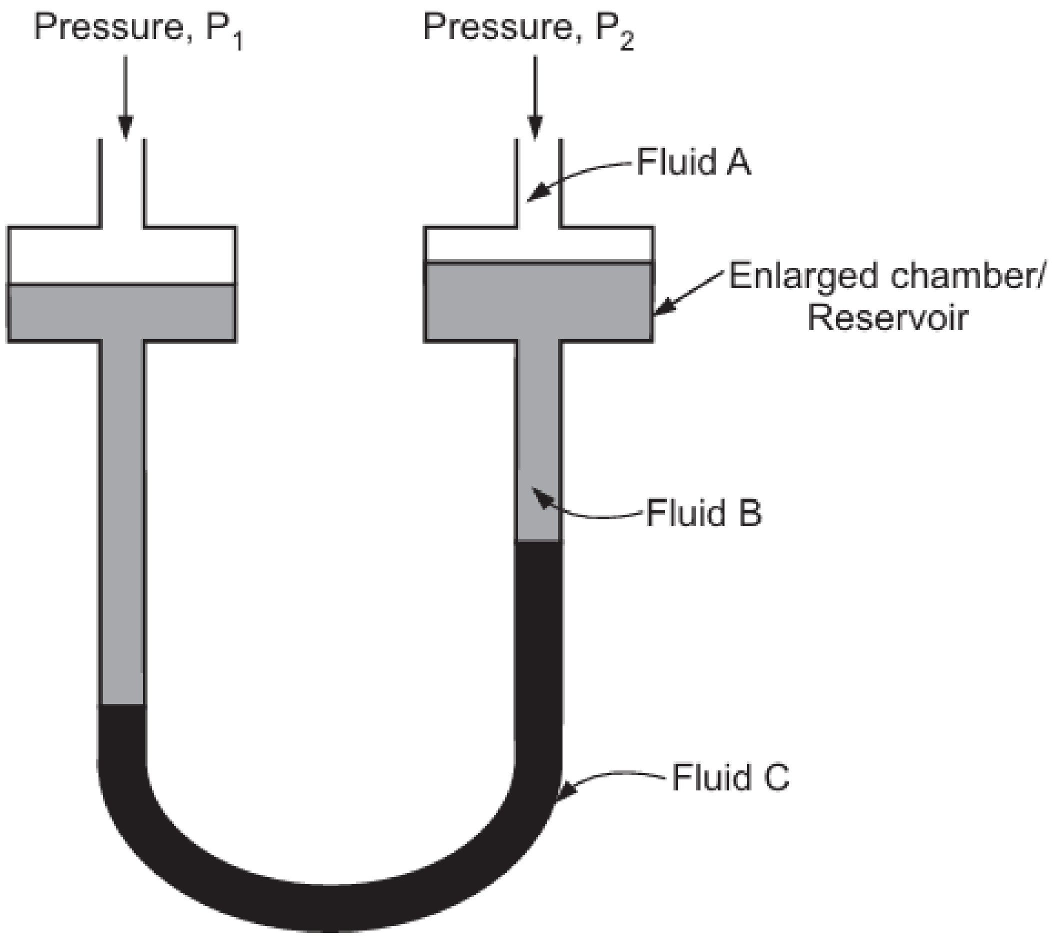
$$\mathbf{(P_1 - P_2) = R \sin \alpha \, (\rho_M - \rho) \, g}$$

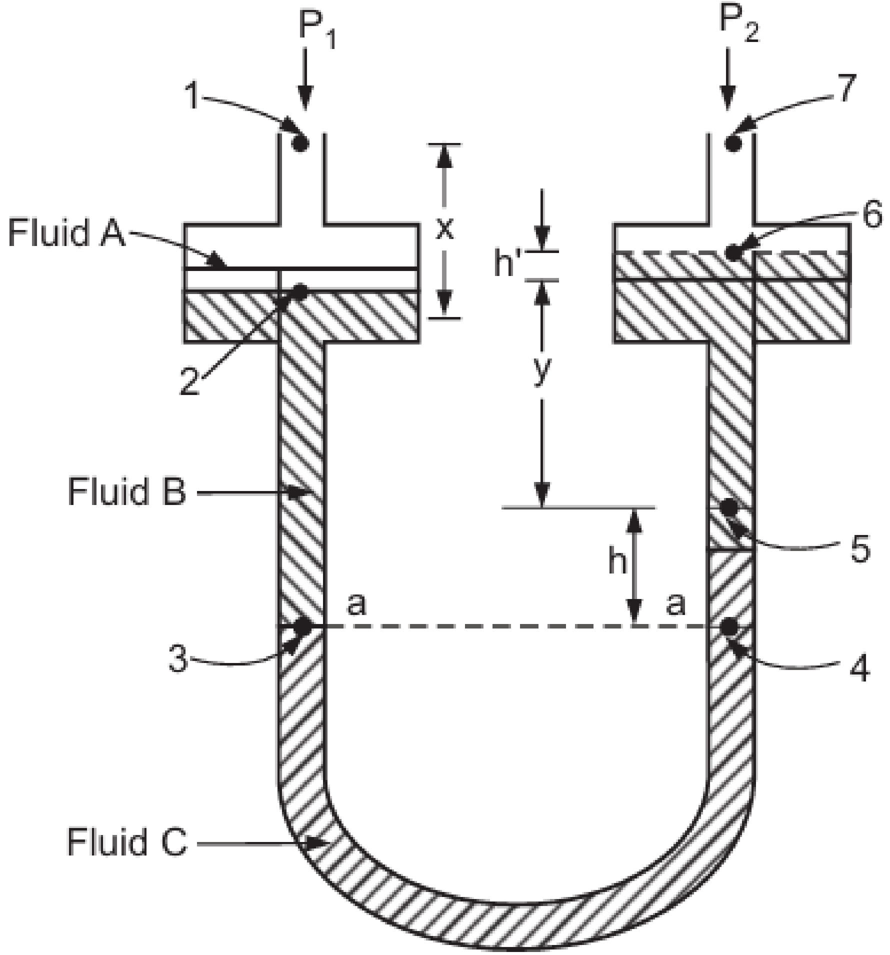
Pressure measurement: Differential Manometer

Differential Manometer / Two Liquid Manometer / Multiplying Gauge

- **Differential manometer is used for the measurement of very small pressure differences or for the measurement of pressure differences with a very high precision.**
- It may often be used for **gases**.
- It consists of a U-tube made of glass. The ends of the tube are connected to two enlarged transparent chambers / reservoirs.
- **The reservoirs at the ends of each arm are of a large cross-section than that of tube.**
- The manometer contains **two manometric liquids of different densities** and these are **immiscible with each other and with the fluid** for which the pressure difference is to be measured.
- The **densities of the manometric fluids are nearly equal** to have a high sensitivity of the manometer. Liquids which give sharp interfaces are commonly used, e.g., **paraffin oil and industrial alcohol, etc.**

Pressure measurement: Differential Manometer





Pressure measurement: Differential Manometer

- Let the flowing fluid be 'A' of density ρ_A and manometric fluids be B and C of densities ρ_B and ρ_C ($\rho_C > \rho_B$), respectively [$\boldsymbol{\rho_A < \rho_B}$ **and** $\boldsymbol{\rho_C}$].
- The pressure difference between two points (1 and 7) can be obtained by writing down pressures at points 1, 2, 3, 4, 5, 6, and 7 and is given by

$$P_1 - P_2 = h' (\rho_B - \rho_A) g + h (\rho_C - \rho_B) g$$

- If the level of liquid in two reservoirs is approximately same, then $h' \approx 0$ and Equation reduces, where h is the difference in level in the two arms/limbs of the manometer.

$$P_1 - P_2 = h (\rho_C - \rho_B) g$$

- When the densities ρ_B and ρ_C are nearly equal [($\rho_C - \rho_B$) small], then very large values of h can be obtained for small pressure differences.

Pressure measurement: Differential Manometer

- Alternately, the pressure at the level a – a in Fig. must be the same in each of the limbs and therefore,

$$P_1 + [x.\rho_A + h'\,\rho_A + y.\rho_B + h.\rho_B]\,g = P_2 + [x.\rho_A + h'\,\rho_B + y\,.\rho_B + h.\rho_C]\,g$$

$$\mathbf{(P_1 - P_2) = h'\,(\rho_B - \rho_A)\,g + h\,(\rho_C - \rho_B)\,g}$$

Working of Manometers

<u>YouTube Links:</u>
https://www.youtube.com/watch?v=9SQ2FPHCZJk

https://www.youtube.com/watch?v=gxrkLkJybnA

Simulation Experiment: https://learncheme.com/quiz-yourself/interactive-self-study-modules/manometry/manometry-simulation/

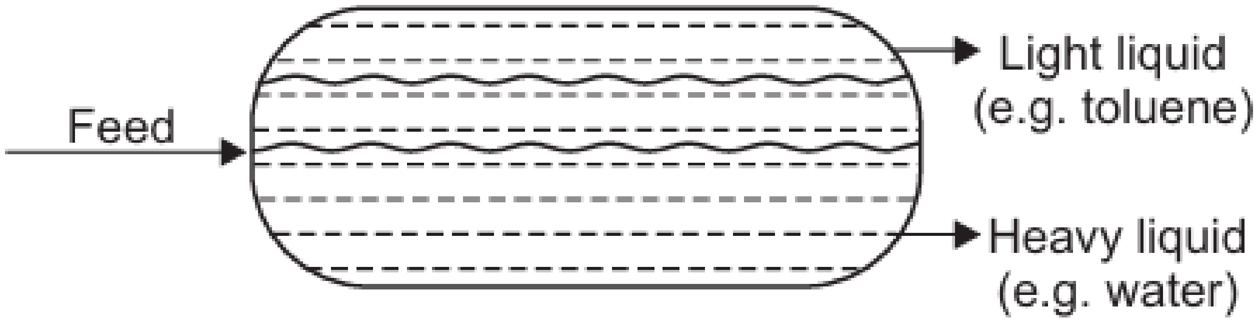
Virtual Lab calculations:
https://fm-pm-iitb.vlabs.ac.in/exp/manometer/simulation.html

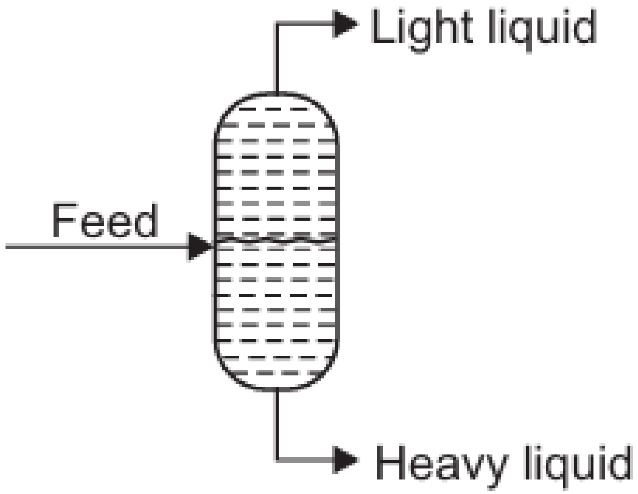
Continuous Gravity Decanter

- **Decantation involves the separation of two immiscible liquids of differing densities from one another.**
- Basically, the difference in densities of two immiscible liquids is responsible for such a separation.
- Decanters used for the separation of two immiscible liquids are : (i) gravity decanter and (ii) centrifugal decanter.
- **Decanters utilize either a gravitational force or a centrifugal force to effect the separation.**
- A **gravity decanter** is used for the separation of two immiscible liquids when the difference between densities of the two liquids is large.
- A **centrifugal decanter** is used for the separation of two immiscible liquids whenever the difference between densities of the two liquids is small.
- The separating force (centrifugal force) in the centrifugal decanter is much larger than the force of gravity.

Continuous Gravity Decanter

- Separation of two immiscible liquids based on the density difference of the phases involved is commonly encountered in the mass transfer operation such as liquid-liquid extraction.
- Feed enters into the decanter at one end, two immiscible liquids flow slowly, separate into two layers based on the density difference, and then finally the separated layers leave the decanter through the overflow lines at the other end.





Dimensionless equations and dimensionless analysis

- **Dimensionless equations: Equations derived directly from the basic laws of the physical sciences consist of terms that have the same dimensions.**
- An equation in which all terms have the same dimensions is called a dimensionally homogeneous equation.
- If a dimensionally homogeneous equation is divided by any one of its terms, then the dimensions of each term cancel and only numerical magnitudes remain, such an equation is called a **dimensionless equation**.
- **Dimensional analysis is a method of correlating the number of variables into a single equation stating clearly an effect.**
- When the value of a given physical quantity is influenced/affected by a number of variables, then it is not possible to determine their individual effects by experimental methods.
- ***In such cases, the problem can be made more easily handled by making use of the method of dimensional analysis wherein the variables are arranged in dimensionless groups which are considerably less than the number of variables.***

Dimensionless numbers

- **Groups of variables having no dimensions are known as dimensionless groups.**
- Several important dimensionless groups have been found by using a method of dimensional analysis.
- The numerical value of a dimensionless group is the same for any consistent set of units used for the variables within the groups.

Examples:

- **Reynolds number** = $N_{Re} = Du\rho/\mu$
- **Prandtl number** = $N_{Pr} = C_P\mu/K$

Dimensional analysis: Rayleigh Method

- If $Q_1, Q_2, \ldots, Q_n$ are 'n' dimensional variables
- Q_1 is the dependent variable
- $Q_2, Q_3, \ldots, Q_n$ are the independent variables
- According to this method of dimensional analysis :

Q_1 varies as $Q_2^a\, Q_3^b$

- The dimensionless groups are obtained by evaluating the powers/exponents so that the relationship among the variables is dimensionally homogeneous.
- Similar/like powers or exponents of the quantities are grouped together in order to get dimensionless groups.

Dimensional analysis: Rayleigh Method

Example:

- The power required P for an agitator depends upon the propeller diameter D, the rotational speed N of the agitator, the liquid density ρ, the viscosity μ and the gravitational acceleration g.
- Find, by a dimensional analysis, the correct representation for the power requirement in terms of dimensionless groups.

Parameter	Units	Dimensions
Power, P	$(N.m)/s = (kg.m^2)/s^2$	$ML^2\,\theta^{-3}$
Diameter, D	m	L
Speed, N	1/s	θ^{-1}
Viscosity, μ	kg/(m.s)	$ML^{-1}\,\theta^{-1}$
Density, ρ	kg/m^3	ML^{-3}
Gravitational acceleration, g	m/s^2	$L\theta^{-2}$

$$P = f\,(D, N, P, \mu, g)$$

$$P = k\, D^a\, N^b\, \rho^c\, \mu^d\, g^e$$

Dimensional analysis: Rayleigh Method

$$P = k\, D^a\, N^b\, \rho^c\, \mu^d\, g^e$$

$$[ML^2\theta^{-3}] = k\, [L]^a\, [\theta^{-1}]^b\, [ML^{-3}]^c\, [ML^{-1}\theta^{-1}]^d\, [L\theta^{-2}]^e$$

- Equating the exponents / indices of M, L and θ

Exponents of M : $1 = c + d$

Exponents of L : $2 = a - 3c - d + e$

Exponents of θ : $-3 = -b - d - 2e$

Solving $a = 5 - 2d - e$

- Substiuting in 1st Equation:

$$P = k\, (D)^{5-2d-e}\, (N)^{3-d-2e}\, (\rho)^{1-d}\, (\mu)^d\, (g)^e$$

$$\boxed{\underbrace{\frac{P}{D^5 N^3 \rho}}_{\text{Power number, } N_{Np}} = k \left(\underbrace{\frac{D^2 N \rho}{\mu}}_{\text{Reynolds number for agitator, } N_{Re}}\right)^{-d} \left(\underbrace{\frac{g}{N^2 D}}_{\text{Froude number, } N_{Fr}}\right)^{e}}$$

Dimensional analysis: Buckingham's π theorem

- **Buckingham's π theorem states that the number of dimensionless groups is equal to the number of variables minus the number of fundamental dimensions.**
- If there are 'n' variables, Q_1, Q_2 …… Q_n affecting a given physical process, the functional relationship between them can be written as

$$f(Q_1, Q_2, ……, Q_n) = 0$$

- If there are 'm' fundamental dimensions (such as M, L, θ, etc.) required to define the dimensions of Q_1, Q_2,..., Q_n, then there will be n – m dimensionless groups (π_1, π_2,.., π_{n-m}).
- The functional relationship between them can be written as

$$f(\pi_1, \pi_2, …… , \pi_{n-m}) = 0$$

- The groups π_1, π_2, etc. must be independent of each other.
- With the help of Buckingham's π theorem, it is possible to obtain the dimensionless groups more easily than by solving the simultaneous equations for the exponents/indices.

Dimensional analysis: Buckingham's π theorem

Procedure :

(a) Select m of the original variables in order to form what is called a recurring set.

(b) The dependent variable, that is, the variable that depends upon (or gets affected by) the other variables, should not be included in the recurring set.

(c) Any set m of the variables may be selected with the following provisions.

(i) Each of the fundamental/basic dimensions must appear in at least one of the m variables.

(ii) Some or all of the variables within the recurring set must not form a dimensionless group.

(d) Take each of the remaining n – m variables on its own and form it into a dimensionless group by combining it with one or more members of the recurring set.

In this manner, the n – m dimensionless groups are formed.

Dimensional analysis: Buckingham's π theorem

- The relationship between the variables affecting the pressure drop for flow of a fluid in a pipe may be written as : f (ΔP, d, l, u, ρ, μ) = 0
- There are six variables and three fundamental dimensions (mass, length and time).
- **Therefore, number of dimensionless groups = 6 – 3 = 3.**
- The recurring set must contain three variables that cannot themselves form a dimensionless group.

(i) l and d cannot be selected as they can be themselves formed into the dimensionless group l/d.

(ii) ΔP, ρ and u cannot be selected as they can be formed into the dimensionless group $\Delta P/\rho u^2$.

(iii) ΔP is the affected variable.

- Therefore, it cannot be included in the recurring set.

Dimensional analysis: Buckingham's π theorem

- **If we select the variables d, u, ρ as the recurring set, then this set fulfils all the above conditions.**
- Dimensionally : $d = L$, $u = L\theta^{-1}$, $\rho = ML^{-3}$
- Each of the dimensions M, L and θ can then be obtained in terms of the variables d, u and ρ.

$$L = D,\ M = \rho\, D^3,\ \theta = Du^{-1}$$

- **Thus, the three dimensionless groups are formed by taking each of the remaining variables ΔP, l and μ in turn.**

(i) Taking ΔP to form one dimensionless group. ΔP has dimensions $ML^{-1}\,\theta^{-2}$

$\Delta P\ M^{-1}\ L\ \theta^{2}$ is therefore dimensionless.

- Therefore, group π_1 is

$$\pi_1 = \Delta P\ (\rho D^3)^{-1}\ (D)\ (Du^{-1})^2 = \Delta P/\rho u^2$$

Dimensional analysis: Buckingham's π theorem

(ii) Taking l : l has dimension L

l .L^{-1} is therefore dimensionless.

- Therefore, group π_2 is

$$\pi_2 = l\,(D)–1 = l/D$$

(iii) Taking μ : μ has dimensions $ML^{-1}\,\theta^{-1}$

$\mu\, M^{-1}\, L\, T$ is therefore dimensionless.

- Therefore, group π_3 is

$$\pi_3 = \mu\,(\rho\, D^3)^{-1}\,(D)\,(Du^{-1}) = \mu / Du\rho$$

$$\Rightarrow \quad f\left(\frac{\Delta P}{\rho\, u^2}, \frac{l}{D}, \frac{\mu}{Du\rho}\right) = 0 \qquad \Rightarrow \quad \frac{\Delta P}{\rho u^2} = f\left(\frac{l}{D}, \frac{Du\rho}{\mu}\right)$$